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Probabilistic Detection of Market Manipulation

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Abstract

Acknowledgements

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Chapter 1

Introduction

1.1 Context and Motivation

Modern securities exchanges organise trading through electronic limit order books. A limit order book (LOB) records all outstanding buy and sell orders at each price level, providing a real-time, public summary of supply and demand. The instantaneous state of the book, defined by the distribution of volume across price levels on both sides, and the recent flow of incoming orders jointly determine short-term price dynamics. Tao et al. [6] model this dependence through a multi-level volume imbalance that aggregates supply and demand across several price levels, showing that imbalance shifts are directly linked to the probability distribution of subsequent price moves. Fabre and Challet [1] extend this line of work by introducing Hawkes-inspired order flow features that capture both the size and the distance of placement of new limit orders, demonstrating on cryptocurrency data that these features predict the conditional distribution of mid-price movements.

Market manipulation exploits this dependence. By artificially altering either the instantaneous book state or the order flow, a manipulator can induce a predictable price response and profit from it. Spoofing, the primary focus of this project, consists of placing large visible limit orders with no intention of execution. These orders shift the perceived supply-demand balance, move the mid-price in the desired direction, and are cancelled once the manipulator's genuine order on the opposite side has been filled. Layering extends spoofing by stacking multiple non-bona-fide orders across several price levels, amplifying the perceived depth on one side of the book [1]. Both strategies distort the volume imbalance: Tao et al. [6] show that the optimal spoofing action shifts the multi-level imbalance away from its equilibrium value, and that comparing the observed imbalance before a market order to its long-run distribution provides a basis for detection.

The regulatory treatment of these strategies differs sharply between asset classes. In regulated equity and derivatives markets, spoofing and layering are explicitly prohibited under the European Market Abuse Regulation (EU MAR) and the US Dodd-Frank Act. In cryptocurrency markets, the absence of equivalent regulation opens the door to manipulative behaviour on a larger scale [1]. Fabre and Challet, working on Level-3 crypto order book data, find that 31% of large orders on the BTC-USD pair could profitably spoof the market over a four-day period. This asymmetry suggests that crypto order book data should contain a substantially higher base rate of manipulative patterns than equity data, a hypothesis that motivates future comparative work.

A binary classifier that labels each event as "manipulative" or "not" is poorly suited to this detection problem for three reasons. First, no ground-truth labels exist: in equity markets,

confirmed manipulation cases are rare and not publicly annotated at the order level. Second, manipulation is not a discrete category but a spectrum of intent, from aggressive but legitimate liquidity provision to unambiguous spoofing. Third, a probabilistic score is more actionable than a binary flag: it allows a surveillance system to rank events by severity and calibrate alert thresholds to a desired false discovery rate. Fabre and Challet [1] address these requirements with a probabilistic neural network that outputs a full predictive distribution over future price moves, from which both an anomaly score (the negative log-likelihood) and an economic spoofing gain estimate can be derived.

1.2 Research Questions

This project investigates the following questions:

1. Can unsupervised probabilistic models trained on order book features detect anomalous order flow in the absence of ground-truth labels?
2. Which order book features carry the most information about anomalous events, and do different detection models agree on the same discriminative features?
3. How should anomaly thresholds be calibrated in a non-stationary, streaming setting when no labelled data are available?
4. Can the economic profitability of a spoofing strategy be estimated directly from model predictions, providing an interpretable measure of manipulation opportunity?
5. Do detection patterns remain stable across different time periods, when models trained on one year are applied to data from other years?

These questions are addressed on the TOTF.PA (TotalEnergies, Euronext Paris) limit order book dataset, covering trading days from 2009, 2010, 2015, and 2017.

1.3 Contributions

The main contributions of this report are:

1. A modular detection pipeline that combines three complementary unsupervised models on the same equity LOB dataset (TOTF.PA, Euronext Paris): a Transformer bottleneck autoencoder inspired by Poutré et al. [3], coupled with a Nyström-approximated One-Class SVM in latent space; a Probabilistic Neural Network (PNN) that predicts skew-normal price move distributions following the framework of Fabre and Challet [1]; and a Probabilistic Robust Autoencoder (PRAE) with learnable per-sample gates [2]. Each model produces a scalar anomaly score per time step; their agreement identifies the highest-confidence anomalies.
2. A feature engineering library of approximately 80 features synthesising contributions from multiple sources: Hawkes-inspired memory features with spatial and temporal decay [1], weighted multi-level volume imbalance [6], event flow and cancellation rapidity [3], multi-level order flow imbalance, and standard microstructural indicators (spread, volatility, elasticity).
3. Application of the spoofing gain framework of Fabre and Challet [1] to equity LOB data. The PNN-predicted skew-normal parameters feed an expected cost model that quantifies the profitability of a hypothetical spoofing strategy at each time step, producing an interpretable economic measure of manipulation opportunity.

4. A systematic comparison of four threshold methods for converting continuous anomaly scores into binary alerts without labels: Peak Over Threshold (POT), Streaming POT (SPOT), Drift-aware Streaming POT (DSPOT) [4], and a rolling Benjamini-Hochberg False Discovery Rate controller (RFDR).
5. Interpretability analysis via Integrated Gradients [5] for the PNN and PRAE, and Grouped Occlusion for the Transformer+OC-SVM, identifying which feature families and which sides of the order book drive each model's anomaly scores.
6. Post-hoc characterisation of detected anomalies through HDBSCAN clustering of the three anomaly scores and microstructural jump analysis (bipower variation, variance ratio, CUSUM), providing an unsupervised taxonomy of anomaly types.
7. Cross-year robustness evaluation: models trained on 2015 data are tested on 2009, 2010, and 2017 data, and models trained on 2017 data are tested on the same periods, assessing whether the pipeline detects consistent patterns across different market regimes.

1.4 Report Outline

Chapter 2 reviews the literature on limit order book microstructure, market manipulation, and the detection methods that this project builds on. Chapter 3 formalises the feature engineering pipeline, the three detection models with their loss functions, the threshold methods, the spoofing gain framework, and the interpretability and post-hoc analysis tools. Chapter 4 describes the TOTF.PA dataset, the preprocessing steps, the sequential training protocol, and the experimental setup. Chapter 5 presents the experimental findings: model diagnostics, anomaly detection rates, threshold comparisons, sensitivity analysis, anomaly clustering, jump analysis, and cross-year robustness. Chapter 6 discusses the limitations of the approach, compares the three detection models, and interprets the detected anomalies in light of known manipulation signatures. Chapter 7 summarises the contributions and outlines directions for future work.

Chapter 2

Background and Related Work

This chapter provides the conceptual vocabulary and critical perspective needed to evaluate the methodological choices presented in Chapter 3. Each section introduces a class of approaches, explains what it is good at, where it struggles, and what assumptions it makes. Formal definitions, model architectures, and training procedures are deferred to Chapter 3.

2.1 Limit Order Books and Market Microstructure

A limit order book records all outstanding buy and sell orders for a security, organised by price and submission time. At time t, the book state consists of L price levels on each side. The bid (buy) side lists prices $p_1^b > p_2^b > \cdots > p_L^b$ with corresponding volumes $v_1^b, v_2^b, \ldots, v_L^b$. The ask (sell) side lists prices $p_1^a < p_2^a < \cdots < p_L^a$ with volumes $v_1^a, v_2^a, \ldots, v_L^a$. The best bid is p_1^b, the best ask is p_1^a, the mid-price is $p_{\text{mid}} = (p_1^a + p_1^b)/2$, and the spread is $s = p_1^a - p_1^b$.

Three event types modify the book state. A *limit order* adds volume at a specified price level, increasing liquidity. A *cancellation* removes a previously placed limit order, reducing liquidity at that level. A *market order* (equivalently, a trade) consumes volume at the best price on the opposite side, removing liquidity and potentially shifting the best price.

The instantaneous distribution of volume across price levels and the temporal sequence of these events jointly determine short-term price dynamics. Volume imbalance, the normalised excess of bid over ask volume at a given level, is a widely used summary statistic of the book state. Tao et al. [6] extend the single-level imbalance to a weighted multi-level measure that aggregates supply and demand across several price levels, and show on TMX (Toronto Stock Exchange) data that this quantity is directly linked to the probability distribution of subsequent price moves. Fabre and Challet [1] complement imbalance with Hawkes-inspired features that capture both the size and the spatial placement depth of new limit orders, demonstrating on cryptocurrency data that these features predict the conditional distribution of mid-price movements.

2.2 Market Manipulation Strategies

Spoofing consists of submitting large, visible limit orders with no intention of execution to create a false impression of supply or demand. The lifecycle of a spoofing event follows a characteristic sequence: (1) the manipulator places one or more large limit orders on one side of the book, shifting the perceived imbalance; (2) the mid-price moves in the direction favoured by the manipulator; (3) the manipulator executes a genuine order on the opposite side at the improved price; (4) the non-bona-fide orders are cancelled. Layering extends spoofing by

stacking multiple orders across several price levels on the same side, amplifying the perceived depth [1]. Quote stuffing is a related strategy in which a large number of orders are rapidly submitted and cancelled to slow down other participants' systems and exploit the resulting latency advantage.

Both spoofing and layering are prohibited under the European Market Abuse Regulation (EU MAR, Article 12) and the US Dodd-Frank Act (Section 747). Detection is challenging because the same order flow patterns can arise from legitimate trading strategies, such as aggressive liquidity provision or inventory management. Tao et al. [6] formalise the detection problem by modelling the imbalance shift caused by a spoof order and comparing the observed pre-trade imbalance distribution to its long-run counterpart via the Wasserstein distance. Fabre and Challet [1] take an economic approach: rather than asking whether an order flow pattern looks abnormal, they ask whether a rational spoofer could profit from the current book state, defining the spoofing gain as the difference in expected execution cost between a scenario where the spoof order is present and one where it is absent.

A binary classifier that labels each event as manipulative or not is poorly suited to this problem. No ground-truth labels exist at the order level; manipulation is a spectrum of intent rather than a discrete category; and a probabilistic score is more actionable than a binary flag because it allows ranking events by severity and calibrating alert thresholds to a desired false positive rate.

2.3 Anomaly Detection in Financial Data

In the absence of ground-truth labels for manipulation, detection must proceed unsupervised. A central difficulty shared by all unsupervised approaches is the *contamination problem*: training data collected from a real market inevitably contains some anomalous events, and their presence can degrade the learned model of normality. This section surveys three families of approaches and discusses their relative strengths and weaknesses for the LOB anomaly detection problem.

2.3.1 Reconstruction-Based Approaches

Autoencoders learn a compressed representation of the training data and reconstruct each input from that representation. When the training set is predominantly normal, the model learns to reconstruct normal patterns well; anomalous inputs produce high reconstruction error because they deviate from the learned structure. This family has several merits: it requires no labels, captures complex multivariate dependencies, and generalises across instruments and time periods without task-specific supervision.

For order book sequences, the choice of encoder architecture matters. Convolutional and recurrent architectures capture local patterns effectively but struggle with long-range dependencies between distant price levels or between events separated by many time steps. The Transformer architecture addresses this limitation through self-attention, which allows each position in the sequence to attend directly to every other position. Multi-head attention enables the model to simultaneously capture different aspects of the book state, such as volume dynamics at the top of the book, spread behaviour, and order arrival patterns deeper in the queue. Poutré et al. [3] demonstrate on NASDAQ LOBSTER data that a Transformer autoencoder combined with a one-class classifier in the latent space outperforms the reconstruction error alone for detecting synthetically injected manipulation patterns (quote stuffing, layering, and pump-and-dump).

A limitation of deterministic autoencoders is that the reconstruction error is a noisy proxy for anomaly severity: it does not distinguish between inputs that are merely unusual and inputs

that fall outside the training distribution entirely. A probabilistic formulation, in which the autoencoder outputs a distribution over reconstructions rather than a point estimate, addresses this limitation by providing an uncertainty signal. Lindenbaum et al. [2] go further: rather than modifying the reconstruction itself, they introduce learnable per-sample gates that allow the training procedure to downweight suspected anomalies, so that the autoencoder focuses on reconstructing clean data even when the training set is contaminated. This makes the model robust to the contamination problem without requiring an explicit data-cleaning step.

2.3.2 Probabilistic Scoring via Synthetic Manipulation

An alternative to asking "does this look abnormal?" is to ask "how much could a rational spoofer gain from this book state?" Fabre and Challet [1] pursue this reframing by training a probabilistic neural network that predicts the full conditional distribution of the next mid-price move. The predicted distribution serves two purposes: its negative log-likelihood provides a general anomaly score (events where the realised price move is unlikely under the model receive high scores), and its parameters feed an expected cost model that quantifies the profitability of a hypothetical spoofing strategy at each time step.

The merits of this approach are substantial. The spoofing gain score has a direct economic interpretation, which makes it more actionable for a market surveillance system than a generic reconstruction error. Training on the predictive task sidesteps the need for labelled manipulation data: the model learns normal price dynamics, and deviations from that model are flagged regardless of their cause. The probabilistic output naturally quantifies uncertainty over the predicted price move, which matters for heavy-tailed financial data where point predictions are unreliable.

The limitations are equally clear. The spoofing gain depends on a model of the spoofer's strategy (order size, placement distance, fee structure), so it may miss manipulation patterns that deviate from that model. The approach was developed and evaluated on cryptocurrency LOB data (BTC-USD and ETH-USD on Coinbase) [1], where the absence of anti-manipulation regulation results in a higher base rate of manipulative behaviour; its effectiveness on regulated equity markets, where manipulation is rarer and subtler, has not been established in the literature.

2.3.3 One-Class and Density-Based Approaches

One-class classification methods learn a tight boundary around the training data in a transformed feature space and flag anything outside that boundary as anomalous. The One-Class SVM (OC-SVM) is the most widely used instance: it finds a hyperplane in a kernel-induced feature space that separates the data from the origin with maximum margin, controlled by a parameter ν that bounds the fraction of training points allowed outside the boundary.

The strengths of the OC-SVM are computational efficiency and theoretical simplicity. It does not require specifying what an anomaly looks like, only what normal data looks like, and the kernel trick allows it to capture non-linear structure in the input space. However, it does not scale well to high-dimensional feature spaces without careful kernel design: in the approximately 80-dimensional feature space typical of LOB anomaly detection, the RBF kernel becomes poorly calibrated and the decision boundary may either collapse or become too loose. Low-rank kernel approximations (such as the Nyström method) partially address this issue by projecting the kernel evaluation into a tractable subspace, enabling GPU-accelerated training. A further limitation is that the OC-SVM does not produce a calibrated probability; its decision function is a signed distance, not a likelihood, which complicates downstream interpretation and threshold calibration.

For the LOB problem, the OC-SVM is most effective when applied not to raw features but to a learned latent representation. Poutré et al. [3] demonstrate this by fitting an OC-SVM in the bottleneck space of a Transformer autoencoder, where the reduced dimensionality and the learned structure of the representation make the kernel boundary more discriminative. This two-stage approach combines the representation power of reconstruction-based methods with the boundary-learning capacity of one-class classification.

2.4 Thresholding Methods for Anomaly Detection

Any continuous anomaly score requires a threshold to produce a detection decision. The naive approach, a fixed percentile of the score distribution, assumes that the distribution is stationary and well-behaved. In financial data, score distributions are heavy-tailed and non-stationary: a fixed percentile will produce too many false positives during volatile periods and miss anomalies during quiet periods. Setting thresholds without labels is therefore a problem in its own right.

Extreme Value Theory (EVT). EVT models the tail of the score distribution rather than the bulk, fitting a Generalised Pareto Distribution to the exceedances above a high initial threshold. This is well-suited to heavy-tailed data because the statistical guarantee applies specifically to extreme quantiles, where fixed-percentile methods are least reliable. Siffer et al. [4] extend the batch EVT estimator to streaming settings with two algorithms: SPOT, which updates the tail model as new peaks arrive, and DSPOT, which adds a drift correction that tracks shifts in the baseline score level over time. The main assumption is that the tail of the score distribution is well-approximated by the GPD, which must be verified empirically for each model output.

False Discovery Rate (FDR) Control. FDR-based procedures, such as the Benjamini-Hochberg method, control the expected proportion of false positives among all flagged events. This is appropriate when anomalies are rare and the cost of false positives is high, because it bounds the discovery error rate rather than the absolute false positive count. The requirement is a set of p-values or a reference null distribution for the anomaly score, which must be estimated from the data when no parametric form is available.

Rolling FDR (RFDR). RFDR adapts the Benjamini-Hochberg procedure to a streaming setting by maintaining a sliding window of recent anomaly scores. Within each window, scores are log-transformed, converted to robust z-scores using the median absolute deviation, and mapped to p-values under a Gaussian assumption; the BH procedure then determines a dynamic threshold that controls the false discovery rate at level α. Because the null distribution is re-estimated at every step from the local window, RFDR tracks non-stationarity in the score distribution without requiring an explicit drift model.

Implicit Model Thresholds. Some detection methods produce a threshold as a byproduct of training rather than as a post-hoc step. The OC-SVM decision boundary, for instance, defines an implicit threshold in the feature space: points with negative decision function values are classified as anomalous. This is both a strength (the threshold is jointly optimised with the model) and a weakness (the threshold is not interpretable independently of the model and cannot be recalibrated without retraining).

2.5 Feature Attribution Methods

A detection score alone is not actionable for a market surveillance analyst: the analyst needs to know which features of the order book drove the score and whether those features align with known manipulation signatures. Attribution for sequential financial data is harder than for static inputs because features at different time steps and price levels are structurally correlated,

and because the underlying distribution shifts over the trading day.

Two broad families of attribution methods are relevant. Gradient-based methods, such as Integrated Gradients [5], accumulate the model's gradient along a path from a baseline input to the actual input, producing per-feature importance scores that are theoretically principled (they satisfy completeness and sensitivity axioms) and computationally tractable. Their limitation is that they require a differentiable end-to-end model, which excludes pipelines where a non-differentiable component (such as an OC-SVM) sits between the learned representation and the final score.

For non-differentiable pipelines, ablation-based methods provide an alternative. Grouped occlusion replaces semantically related groups of features (all bid-side features, all Hawkes features, all features at a given price level) with a baseline value and measures the resulting change in the anomaly score. This approach is model-agnostic and handles the structured correlation among LOB features by operating at the group level rather than the individual feature level. The trade-off is lower resolution: it identifies which feature families matter, not which individual features within a family are most important.

Chapter 3

Methodology

3.1 Feature Engineering

The raw limit order book data for TOTF.PA (TotalEnergies, Euronext Paris) records the state of the book at each update event during continuous trading hours (09:00–17:30 CET). Each row contains an Excel serial timestamp t (column `xltime`, where the fractional part encodes the time of day) and the price and volume at $L = 10$ levels on each side: $p_1^b, V_1^b, \ldots, p_{10}^b, V_{10}^b$ for the bid and $p_1^a, V_1^a, \ldots, p_{10}^a, V_{10}^a$ for the ask, giving 41 columns per row. The data is event-driven: a new row is recorded whenever a limit order, cancellation, or trade modifies the visible book state, so the inter-event spacing is irregular.

Feeding the 40 raw price-volume columns directly to a model is wasteful and fragile. Prices at adjacent levels are highly correlated, volumes are non-stationary, and the absolute price level carries no information about short-term dynamics. Engineered features reduce the input dimensionality to $d \approx 80$ informative quantities that are more stationary, more interpretable, and encode the domain knowledge needed to distinguish normal from anomalous order flow.

The features are organised into seven groups: (1) order book imbalance, capturing supply-demand asymmetry across depth levels; (2) price dynamics and microstructure, summarising price movements and their relationship to volume; (3) liquidity and elasticity, measuring the cost and shape of the book; (4) volatility proxies; (5) event flow and rapidity, characterising the intensity and type of order arrivals; (6) Hawkes-inspired memory features, capturing self-exciting patterns in order flow with spatial and temporal decay; and (7) order flow imbalance, measuring signed directional pressure from volume changes conditioned on price movements. Preprocessing and scaling are described in Section 3.1.8.

3.1.1 Order Book Imbalance

Imbalance features quantify the asymmetry between supply and demand at various depths of the book. Spoofing and layering distort this asymmetry by injecting volume on one side; imbalance measures are therefore among the most direct indicators of manipulative intent.

The Level-1 imbalance compares bid and ask volume at the best quotes:

$$I_t^{(1)} = \frac{V_{1,t}^b - V_{1,t}^a}{V_{1,t}^b + V_{1,t}^a}. \tag{3.1}$$

Values near $+1$ indicate strong buy-side pressure; values near -1 indicate sell-side pressure.

The Level-5 imbalance aggregates volume across the top five levels on each side:

$$I_t^{(5)} = \frac{\sum_{i=1}^{5} V_{i,t}^b - \sum_{i=1}^{5} V_{i,t}^a}{\sum_{i=1}^{5} V_{i,t}^b + \sum_{i=1}^{5} V_{i,t}^a}. \tag{3.2}$$

A divergence between $I_t^{(1)}$ and $I_t^{(5)}$ suggests that volume is concentrated away from the best quotes, a pattern consistent with layering.

To weight deeper levels more flexibly, we follow Tao et al. [6] and define a weighted multi-level imbalance over $L = 5$ levels with weight vector $\mathbf{w} = (w_1, \ldots, w_5)$:

$$I_t^{(\mathbf{w})} = \frac{\sum_{i=1}^{5} w_i V_{i,t}^b}{\sum_{i=1}^{5} w_i V_{i,t}^b + \sum_{i=1}^{5} w_i V_{i,t}^a}. \tag{3.3}$$

This quantity lies in $[0, 1]$; values above 0.5 indicate net bid-side weight. Three weight configurations are used: *decreasing* $\mathbf{w} = (0.1, 0.1, 0.2, 0.2, 0.4)$, which emphasises deeper levels where spoof orders are less likely to be executed; *increasing* $\mathbf{w} = (0.4, 0.2, 0.2, 0.1, 0.1)$, which emphasises the best quotes; and *constant* $\mathbf{w} = (0.2, 0.2, 0.2, 0.2, 0.2)$, which weights all five levels equally.

3.1.2 Price Dynamics and Microstructure

This group captures the immediate state of the price process and its first- and second-order dynamics.

The mid-price and bid-ask spread are:

$$p_{\text{mid},t} = \frac{p_{1,t}^a + p_{1,t}^b}{2}, \qquad s_t = p_{1,t}^a - p_{1,t}^b. \tag{3.4}$$

The micro-price deviation measures how far the L1-imbalance-adjusted fair value lies from the mid-price:

$$\delta_t^{\text{micro}} = I_t^{(1)} \cdot \frac{s_t}{2}. \tag{3.5}$$

A large deviation suggests that the balance of volume at the best quotes is pulling the fair value away from the geometric centre of the spread.

The volume depth ratio measures the concentration of volume away from the best quote on each side $s \in \{b, a\}$:

$$D_t^s = \frac{\sum_{i=2}^{5} V_{i,t}^s}{V_{1,t}^s}. \tag{3.6}$$

High values indicate that most liquidity sits behind the best quote, a signature of layering where large volumes are placed at deeper levels.

Log returns, velocity, and acceleration of the mid-price are:

$$r_t = \ln\left(\frac{p_{\text{mid},t}}{p_{\text{mid},t-1}}\right), \quad v_t = p_{\text{mid},t} - p_{\text{mid},t-1}, \quad a_t = v_t - v_{t-1}. \tag{3.7}$$

The net order flow at Level 1 isolates volume changes that occur at a fixed price, filtering out changes caused by price-level shifts:

$$F_t^s = \Delta V_{1,t}^s \cdot \mathbf{1}\left[p_{1,t}^s = p_{1,t-1}^s\right], \qquad s \in \{b, a\}, \tag{3.8}$$

where $\Delta V_{1,t}^s = V_{1,t}^s - V_{1,t-1}^s$. A large positive F_t^b followed by a rapid cancellation is a textbook spoofing pattern.

Volume deltas $\Delta V_{1,t}^b$ and $\Delta V_{1,t}^a$ at the best quotes are also retained as raw features, along with the rolling standard deviation of the mid-price over a window of $W = 50$ events.

3.1.3 Liquidity and Elasticity

These features describe how much it costs to execute a given order size and how steeply the book's price schedule rises with volume.

The sweep-to-fill cost is the volume-weighted average price paid to execute a target order of size $Q = 1{,}000$ units by walking through $L = 10$ levels. Denoting by $C_t^s(Q)$ the resulting average execution price on side s, the reported features are the excess costs relative to the mid-price:

$$\mathrm{SC}_t^a = C_t^a(Q) - p_{\mathrm{mid},t}, \qquad \mathrm{SC}_t^b = p_{\mathrm{mid},t} - C_t^b(Q). \tag{3.9}$$

Both quantities are non-negative under normal conditions. Large values indicate thin liquidity: a spoofing episode that removes genuine orders from the book can cause a spike in sweep cost.

The book slope (elasticity) measures the sensitivity of price to cumulative volume. For each side s, a linear regression of price on log cumulative volume across the first $D = 5$ levels gives:

$$p_{i,t}^s \approx \alpha_t^s + \beta_t^s \ln\left(\sum_{j=1}^{i} V_{j,t}^s + 1\right), \qquad i = 1, \ldots, D, \tag{3.10}$$

and the feature is the absolute slope $|\beta_t^s|$. A steep slope (large $|\beta_t^s|$) indicates low liquidity, where a small volume would cause a large price change; a flat slope indicates a deep, resilient book.

3.1.4 Volatility

Local volatility features capture the recent variability of the price process and of event timing.

Rolling standard deviations of log returns are computed over two horizons:

$$\sigma_t^{(W)} = \sqrt{\frac{1}{W-1} \sum_{k=0}^{W-1} (r_{t-k} - \bar{r}_t)^2}, \qquad W \in \{50, 100\}. \tag{3.11}$$

The normalised price range over a window of $W = 50$ events measures extreme movements:

$$R_t = \frac{\max_{k \in [t-W+1,t]} p_{\mathrm{mid},k} - \min_{k \in [t-W+1,t]} p_{\mathrm{mid},k}}{p_{\mathrm{mid},t}}. \tag{3.12}$$

A sudden spike in R_t without a corresponding increase in $\sigma_t^{(50)}$ may indicate a transient price dislocation caused by manipulation.

The absolute velocity $|r_t|$ captures the instantaneous magnitude of price movement without directional information.

The inter-event time $\Delta t_t = t - t_{t-1}$ (in fractions of a day, following the Excel `xltime` convention) and its logarithm $\ln(\Delta t_t + \epsilon)$, with $\epsilon = 10^{-6}$, are included to provide the models with information about event timing. Abnormally short inter-event times can signal quote-stuffing activity.

3.1.5 Event Flow and Rapidity

Following Poutré et al. [3], each LOB update is classified as either a cancellation or a trade based on observable price and volume changes. A cancellation on side s is identified when the best price is unchanged and volume decreases ($p^s_{1,t} = p^s_{1,t-1}$ and $\Delta V^s_{1,t} < 0$). A trade on the bid (ask) side is identified when the opposite ask (bid) price changes, indicating that a market order has consumed liquidity.

The event size features record the magnitude of each event type:

$$\text{Size}_t^{\text{cancel},s} = |\Delta V^s_{1,t}| \cdot \mathbf{1}[\text{cancel}^s_t], \qquad \text{Size}_t^{\text{trade},s} = V^s_{1,t-1} \cdot \mathbf{1}[\text{trade}^s_t]. \tag{3.13}$$

Each of these four quantities, along with the raw Level-1 volume, is smoothed with a simple moving average over $W_{\text{SMA}} = 10$ events.

Rapidity features measure the density of events per unit of calendar time:

$$\rho_t^{\text{type},s} = \frac{\mathbf{1}[\text{type}^s_t]}{\Delta t_t}, \qquad \text{type} \in \{\text{cancel}, \text{trade}\}, \quad s \in \{b, a\}. \tag{3.14}$$

High cancellation rapidity on one side of the book is a signature of quote stuffing, where a manipulator floods the exchange with rapid submissions and cancellations.

Bid- and ask-side log returns ($\ln(p^s_{1,t}/p^s_{1,t-1})$ for $s \in \{b, a\}$) and price speed (log return divided by Δt_t) complete this group.

3.1.6 Hawkes-Inspired Memory Features

Following Fabre and Challet [1], we define order flow variables inspired by the self-exciting structure of Hawkes processes. Let $(N_t^{L,b})_{t\geq 0}$ and $(N_t^{L,a})_{t\geq 0}$ denote the point processes counting the arrivals of buy and sell limit orders, and let $(N_t^{M,b})_{t\geq 0}$ and $(N_t^{M,a})_{t\geq 0}$ denote the point processes of sell and buy marketable orders (i.e. orders that cause transactions on the bid and ask sides respectively). Each limit order carries two marks: a volume $v_t^{L,s} > 0$ and a distance of placement $\delta_t^{L,s} > 0$ from the best price, for $s \in \{b, a\}$. Each marketable order carries a volume mark $v_t^{M,s} > 0$.

The limit-order and marketable-order flow variables are defined as

$$L_t^s(\phi) = \int_{(-\infty,t]} \phi(t-u,\, v_u^{L,s},\, \delta_u^{L,s})\, dN_u^{L,s}, \qquad s \in \{b, a\}, \tag{3.15}$$

$$M_t^s(\psi) = \int_{(-\infty,t]} \psi(t-u,\, v_u^{M,s})\, dN_u^{M,s}, \qquad s \in \{b, a\}, \tag{3.16}$$

where ϕ is a positive kernel that weights each past limit order by its elapsed time, volume, and distance of placement, and ψ is a positive kernel that weights each past marketable order by its elapsed time and volume.

The kernels take exponential form with a linear volume function $f(v) = v$:

$$\phi_{\beta,\eta}(t, v, \delta) = e^{-\beta t}\, v\, e^{-\eta\delta}, \qquad \psi_\beta(t, v) = e^{-\beta t}\, v, \tag{3.17}$$

where $\beta > 0$ controls the temporal decay rate and $\eta \geq 0$ controls the spatial decay rate. The exponential time kernel is chosen for two reasons: it satisfies the Markov property, which permits efficient recursive updates in a streaming setting, and combining multiple time scales β approximates the slowly decaying empirical memory of order flow across several decades [1].

Two families of kernels are constructed by varying β over a finite set $\boldsymbol{\beta}$ and η over a finite set $\boldsymbol{\eta}$:

$$\boldsymbol{\phi} = \{\phi_{\beta,\eta} : \beta \in \boldsymbol{\beta},\, \eta \in \boldsymbol{\eta}\}, \qquad \boldsymbol{\psi} = \{\psi_\beta : \beta \in \boldsymbol{\beta}\}. \tag{3.18}$$

The corresponding feature sets at time t are

$$\mathcal{L}_t = \big\{(L_t^b(\phi),\, L_t^a(\phi)) : \phi \in \boldsymbol{\phi}\big\}, \qquad \mathcal{M}_t = \big\{(M_t^b(\psi),\, M_t^a(\psi)) : \psi \in \boldsymbol{\psi}\big\}. \tag{3.19}$$

Each element of $\mathcal{L}_t$ captures limit-order flow at a specific combination of time and distance scales; each element of $\mathcal{M}_t$ captures marketable-order flow at a specific time scale. Small η values give nearly equal weight to orders at all depths; large η values concentrate attention on orders near the best quotes, so that distant spoof orders receive negligible weight.

Adaptation to Level-2 data. Fabre and Challet [1] work with Level-3 data, where each order event is individually observed with its volume and distance of placement. Our dataset provides Level-2 snapshots (aggregate volume at each of $L = 10$ price levels), so the integrals in Equations (3.15)-(3.16) are approximated as follows. At each event t, the limit-order volume at level i on side s is inferred from positive volume changes at unchanged price levels: $[\Delta V_{i,t}^s]^+ \cdot \mathbf{1}[p_{i,t}^s = p_{i,t-1}^s]$. The distance mark $\delta_{i,t}^{L,s}$ is set to $c \cdot |p_{i,t}^s - p_{\text{mid},t}|$, where $c = 10^4$ converts Euro-denominated distances to a dimensionless quantity. Marketable-order flow is inferred from volume decreases at changing price levels. The continuous-time exponential decay $e^{-\beta t}$ is replaced by a discrete EWMA with half-life parameter β (measured in events), applied to each side and kernel configuration independently.

Parameter grid. The time scales are $\boldsymbol{\beta} = \{10, 100, 1{,}000\}$ and the distance scales are $\boldsymbol{\eta} = \{0.001, 0.1, 1.0, 10.0\}$. With two sides of the book, $|\boldsymbol{\beta}| \times |\boldsymbol{\eta}| \times 2 = 24$ limit-order flow features form $\mathcal{L}_t$, and $|\boldsymbol{\beta}| \times 2 = 6$ marketable-order flow features form $\mathcal{M}_t$.

Short- and long-horizon flows. In addition to the kernel-weighted features above, limit-order and marketable-order flows at the best level (Level 1) are smoothed at two fixed half-lives ($h_s = 5$ and $h_\ell = 50$ events) for the limit component and at $h_s = 5$ for the market component, yielding six features: $L_t^b(h_s)$, $L_t^a(h_s)$, $M_t^b(h_s)$, $M_t^a(h_s)$, $L_t^b(h_\ell)$, $L_t^a(h_\ell)$. These correspond to the special case $\eta = 0$ (no spatial decay) with a single time scale, and capture the short- and long-term memory of order flow at the top of the book.

Deep order insertions. Volume increases at Level 5 are tracked separately to capture layering activity deep in the book. The positive volume change at Level 5, smoothed with an EWMA of half-life $h_d = 10$, produces two additional features (bid and ask).

3.1.7 Order Flow Imbalance

Order flow imbalance (OFI) measures the net signed change in supply and demand at each price level, conditioned on whether the price at that level has moved. Unlike the imbalance features of Section 3.1.1, which compare static snapshots of volume, OFI captures the directional pressure exerted by incoming orders.

At each level $i \in \{1, \ldots, 5\}$, the bid-side flow is:

$$\phi_{i,t}^b = \begin{cases} V_{i,t}^b & \text{if } p_{i,t}^b > p_{i,t-1}^b, \\ V_{i,t}^b - V_{i,t-1}^b & \text{if } p_{i,t}^b = p_{i,t-1}^b, \\ -V_{i,t-1}^b & \text{if } p_{i,t}^b < p_{i,t-1}^b, \end{cases} \tag{3.20}$$

and the ask-side flow $\phi_{i,t}^a$ follows the symmetric convention (positive when the ask price improves, i.e. decreases). The OFI at level i is

$$\text{OFI}_{i,t} = \phi_{i,t}^b - \phi_{i,t}^a. \tag{3.21}$$

Positive values indicate net buying pressure; negative values indicate net selling pressure. Computing OFI at five separate levels allows the models to distinguish pressure at the best quotes from pressure deeper in the book, which is relevant for detecting layering.

3.1.8 Preprocessing and Scaling

Before the feature matrix enters any model, four cleaning steps are applied in sequence. First, the initial $W_{\text{warmup}} = 3{,}000$ rows of each trading day are discarded to allow all EWMA and rolling-window features to stabilise. Second, infinite values (arising from division by zero in rare edge cases) are replaced by zero. Third, each feature column is clipped at its 0.1% and 99.9% quantiles to remove extreme outliers while preserving the tail structure needed for anomaly detection. Fourth, any column with standard deviation below 10^{-9} is dropped as uninformative.

Three scaling methods are available, selected per experiment. *MinMax* scaling maps each feature to $[0, 1]$. *Standard* scaling centres each feature to zero mean and unit variance. *Quantile* scaling first applies a quantile transformation (mapping the marginal distribution to a standard Gaussian) and then applies standard scaling; this two-stage procedure is the most robust to the heavy tails typical of LOB features. In all cases, the scaler is fit on the training partition only and then applied to validation and test data, preventing information leakage across splits.

3.2 Detection Models

3.2.1 Transformer Bottleneck Autoencoder + One-Class SVM

Nyström Approximation and SGD Training.

3.2.2 Probabilistic Neural Network

3.2.3 Probabilistic Robust Autoencoder

Lambda Tuning.

3.3 Anomaly Thresholding

3.3.1 Peak Over Threshold

3.3.2 SPOT and DSPOT

3.3.3 Rolling False Discovery Rate

3.4 Spoofing Gain Estimation

3.5 Interpretability

3.5.1 Integrated Gradients

3.5.2 Grouped Occlusion

3.6 Post-Hoc Analysis

3.6.1 Anomaly Clustering

3.6.2 Jump Analysis

Chapter 4

Data and Experimental Setup

4.1 Dataset Description

4.2 Preprocessing Pipeline

4.3 Training Protocol

4.3.1 Model Hyperparameters

4.4 Test Protocol

4.5 Compute Environment

Chapter 5

Results

5.1 Model Diagnostics

5.1.1 Transformer + OC-SVM

5.1.2 PNN

5.1.3 PRAE

5.2 Anomaly Detection on Test Data

5.2.1 Anomaly Rates by Model and Time Period

5.2.2 Score Time Series

5.2.3 Model Agreement

5.3 Threshold Comparison

5.4 Spoofing Gain Analysis

5.5 Sensitivity and Feature Attribution

5.5.1 Integrated Gradients: PNN and PRAE

5.5.2 Grouped Occlusion: Transformer + OC-SVM

5.5.3 Cross-Model Feature Importance

5.6 Anomaly Clustering

5.7 Jump Analysis

5.8 Cross-Year Robustness

Chapter 6

Discussion

6.1 Limitations

6.2 Comparison of Detection Approaches

6.3 Interpretation of Detected Anomalies

Chapter 7

Conclusion

7.1 Summary of Contributions

7.2 Future Work

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Appendix A

Feature Catalogue

Appendix B

Hyperparameter Tables

Appendix C

Additional Figures

C.1 Score Distributions per Test Day

C.2 LOB Snapshots for Selected Anomalies

C.3 Feature Evolution Around Anomalies

Appendix D

Threshold Method Details

Appendix E

Spoofing Gain Derivation